Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
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SKIT COLLEGE
Faculty of Applied Science and Engineering
ME-201– QUIZ #3

Discrete Random Variables & Probability Distributions

Instructor: Prof. CM sir 30 AUGUST 2019

Name: _____
Roll Number: _____

This exam contains 7 pages (including this cover page) and 4 questions. Total of points is 0. Good luck and Happy reading work!

Distribution of Marks

1. In a particular game, a fair die is tossed. If the number of spots showing is either 4 or 5 you win $1, if the number of spots showing is 6 you win 4$, and if the number of spots showing is 1, 2, or 3 you win nothing. Let X be the amount that you win. Which of the following is the expected value of X?
   (a) 1.00
   (b) 2.50
   (c) 4.00
   (d) 6.00

2. The weight of written reports produced in a certain department has a Normal distribution with mean 60 g and standard deviation 12 g. The probability that the next report will weigh less than 45 g is
   (a) 0.1056
   (b) 0.3944
   (c) 0.1045
   (d) 0.8944

3. A small store keeps track of the number X of customers that make a purchase during the first hour that the store is open each day. Based on the records, X has the following probability distribution. The standard deviation of the number of customers that make a purchase during the first hour that the store is open is

   (a) (4 points) $P(X = 1)$

   (b) (3 points) $P(X \geq 4)$

4. A reservation service employs five information operators who receive requests for information independently of one another, each according to a Poisson process with rate $\mu = 2$ per minute.
   (a) (4 points) What is the probability that during a given 1-min period, the first operator receives no requests?
   (b) (4 points) What is the probability that during a given 1-min period, exactly four of the five operators receive no requests?(*Hint*: treat either as a binomial process of 5 trials with 4 successes or consider 5 combinations of Poisson processes, e.g. only 1st operation receives a request or only 2nd operation receives a request and so on)

Probability mass/distribution functions

Binomial Distribution

$$f(x;n,p) = b(x;np) = nxp^x(1-p)^{n-x}$$

$$\mu = E(x) = np$$

$$\sigma_x^2 = np(1-p)$$

Hypergeometric Distribution

$$P(X = x) = h(x;n,M,N) = \frac{MxN - Mn - x}{Nn}$$

$$\mu = E(X) = \frac{nM}{N}$$

$$\sigma_x^2 = n\frac{M}{N}\frac{N-M}{N}\frac{N-n}{N-1}$$

Poisson Distribution

$$P(x;\mu) = e^{-\mu}\frac{\mu^x}{x!}$$

$$E(X) = Var(X) = \mu$$

This page is intentionally left blank to accommodate work that wouldn't fit elsewhere and/or scratch work.